DGA Classifier from Domain Names: Family-First or Direct Binary?

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1 Problem and Motivation

Malware uses Domain Generation Algorithms (DGAs) to generate many domains from a shared seed, so infected hosts can reach the attacker without a hard-coded address that defenders could block [12]. Real detection also uses network signals such as the NXDOMAIN rate and query periodicity [1]; this project focuses on detecting DGAs from the domain name alone. The difficulty is that "DGA" is not one pattern: some families produce random strings (`gqspwpsurdun.ru`, CryptoLocker) or hex hashes (`2c3f1e3326cca35b757ec704b2413a30.org`, Bamital), while others join dictionary words (`kneemanualgirl.com`, `whetherspread.net`) and look much closer to legitimate names.

Question. When the final decision is DGA vs. legitimate, should a classifier first learn the individual DGA families (each family + legitimate, with the prediction collapsed to DGA/legitimate) or learn DGA vs. legitimate directly?

Why it matters. The answer is for teams that build name-based DGA filters, for example in DNS resolvers or security products, where a model must judge a single query before any connection is made. If family-first training wins, family labels are worth collecting even when only a binary decision is needed, and simpler models may become accurate enough to deploy. If it does not, the simpler binary setup is enough.

Goal. For each model we evaluate, measure the difference in macro-F1 on the DGA/legitimate decision between family-first and direct binary training, on the same test set and averaged over 5 random seeds. We count family-first as better when its mean macro-F1 is higher by more than the standard deviation across seeds.

2 Impact

If it succeeds. Teams that build name-based DGA filters get a clear rule for how to train them: if family-first wins, it is worth collecting and maintaining family labels even when only a DGA/legitimate verdict is shipped; if both setups tie, the cheaper binary labels are enough. The per-family results also show which kinds of DGA (random-string or dictionary-based) a name-only detector can and cannot catch, which tells analysts where to expect blind spots.

If it fails. If neither setup separates some families from legitimate domains, in particular dictionary-based ones, that is evidence that the name alone does not carry enough signal for them. Detection would then need network traffic features, such as the NXDOMAIN rate, the periodicity of queries, and how many distinct, previously unseen domains a single host queries [1]. Knowing which families need these signals is itself a useful result.

Risks of using the result.

- *Errors do not cost the same.* A missed DGA domain (false negative) leaves an infected host talking to its attacker. A legitimate domain flagged as DGA (false positive) blocks a real service and adds to alert fatigue. Since almost all real DNS traffic is legitimate, even a small false-positive rate produces many false alarms.
- *Bias in the data.* Legitimate examples come from a popularity ranking (Tranco [8]), which over-represents popular sites. Legitimate but unusual names, such as non-English, short or machine-generated CDN hostnames, may be flagged unfairly. Families with fewer than 300 known domains are left out of our data, so our results say nothing about them.
- *Limited validity.* We test only families seen in training. Attackers can change their algorithms, so results do not guarantee detection of new or adversarial DGAs.

3 Related Work

Woodbridge et al. [16] introduced deep learning for DGA detection with a character-level LSTM trained on Alexa domains and 30 DGA families. They built a binary model and a separate family model: the binary model reached an AUC of 0.9993, but the family model's macro-F1 was only 0.54, rising to 0.96 when similar families were grouped. We adopt macro-F1, and their grouping result suggests that the labels matter. They never collapsed the family model back into a DGA/legitimate detector, which is the comparison we make.

Gregório et al. [6] classified 61 DGArchive families with a character-level CNN, reaching an average F1 of 84%, with some families at 0%. Their model has no legitimate class, so it cannot be used as a detector. We reuse this CNN, add the legitimate class, and test whether confusions between DGA families still matter once predictions are collapsed to DGA/legitimate.

Torrealba Aravena et al. [14] proposed DeepDGA, which combines a character-level BiLSTM with word embeddings for binary detection on 25 families. It exceeds 0.97 precision on dictionary-based DGAs, where feature-based models stay below 0.89. We take dictionary DGAs as the hardest case and report them separately. They close that gap with a bigger architecture; we test whether richer labels help simpler models.

Shahzad et al. [13] compared LSTM, BiLSTM and GRU for binary detection on over two million domains. The LSTM reached 87% accuracy and an AUC of about 0.98, with little difference between the three. We follow their preprocessing (removing duplicates after stripping the TLD) and their 80/10/10 train/validation/test split. Since changing the architecture barely mattered, we change the training labels instead, and report per-family results, which they do not.

Dahal and Kim [3] compressed character-count vectors with an autoencoder and classified them with a neural network or an SVM on 11 DGA families. They reached 88% accuracy, which fell to 78% on families not seen in training. We use character counts, as they do, as a simple input for our baseline. They target unseen families, while we focus on known ones; their drop shows the limit of our stratified split.

4 Data

Sources. We combine two DGA sources and one legitimate source into one table that gives, for each domain, its label (DGA/legitimate) and its family.

- *DGA (main):* "Legitimate domains and DGA categorized morphologically and by families" by Gregório and Cansian, Mendeley Data v1, 2025, `https://data.mendeley.com/datasets/nhvyvytn2h/1`, license CC BY 4.0 [5]. We use its 3,090,661 DGA domains, collected from the DGArchive feed [12] (28 Sep–28 Oct 2025) and labelled with 160 families.
- *DGA (extra):* Kaggle "DGA dataset" by C. Givre (v1, 2021, `https://www.kaggle.com/datasets/gtkcyber/dga-dataset`, license not stated, used for academic research only) [4]. We use its 80,000 DGA domains (7 family labels).
- *Legitimate:* the Tranco top 1M list [8] (list `64VVX`, `https://tranco-list.eu/list/64VVX`), which ranks pay-level domains by combining the rankings of five providers (Chrome UX Report, Farsight, Majestic, Cloudflare Radar, Cisco Umbrella) over 17 Sep–16 Oct 2026.

The legitimate parts of the two DGA datasets (Majestic and Alexa) are not used, so all legitimate domains come from one documented ranking.

Size, features and target. The dataset we use has 408,990 domains: 204,495 legitimate and 204,495 DGA from 97 families, with 300 to 3,000 domains per family (49 families have 3,000), so the largest family is at most 10 times the smallest (Fig. 1, Table 1). The only model input is the domain name without its TLD (1 to 89 characters; a–z, 0–9, '-' and '.'); any numerical features are derived from it. The targets are the binary label (DGA/legitimate) and the family (`legit` or one of the 97 DGA families). Data were loaded with pandas [9] and plotted with Matplotlib [7].

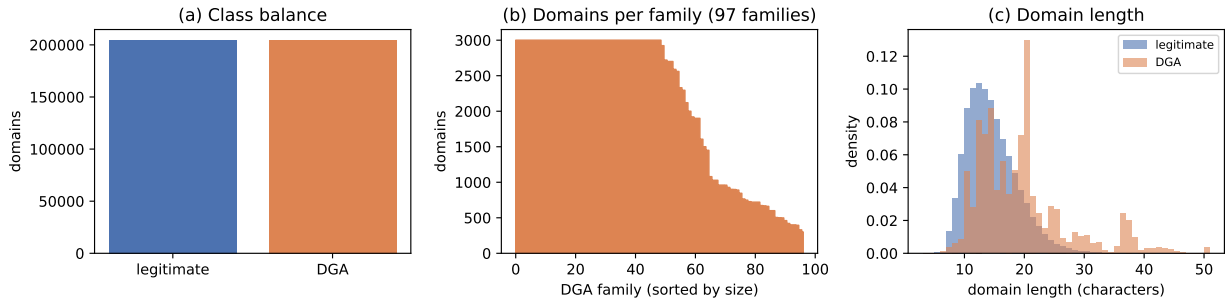


Figure 1: (a) Number of legitimate and DGA domains; (b) number of domains in each of the 97 DGA families; (c) domain length of legitimate and DGA domains.

Table 1: Statistics per class. Entropy is the Shannon entropy (bits) of the characters before the TLD; digit ratio is the share of digits in the same part.

Class	Domains	Families	Mean length	Entropy	Digit ratio	TLDs	Top TLD
Legitimate	204,495	–	14.1	2.85	0.038	625	.com (47%)
DGA	204,495	97	19.2	3.18	0.133	230	.com (28%)

Missing values, noise and biases. There are no empty cells. We found:

- *Label conflicts:* 330 domains are in both Tranco and a DGA list. Some are short DGA names that coincide with real sites (e.g. `qsnatch` generates 3–4 letter names); others are clearly generated (e.g. `pywolwnvd.biz`, family `m0yv`) and likely entered Tranco through the DNS traffic of infected machines. We drop all 330.
- *Noise:* 141 duplicate DGA domains, 6 Tranco wildcard entries (e.g. `_wildcard_.ph`) and one domain containing a line break. The Kaggle `domain` column is truncated at hyphens for some rows, so we use its `host` column.

- *Family names:* no DGA domain appears in both DGA sources (they are years apart), so Kaggle labels were matched by domain shape: `newgoz` and `gameoverdga` → `gameover`, `goz` → the GameOver P2P family, `nivdort` kept as a new family.
- *Biases:* Tranco favours popular sites; each DGA family uses few TLDs (median 3, and 33 families use a single one, vs. 625 for legitimate domains), so the TLD alone would be a shortcut, which is why we remove it; DGA data cover two short periods (around 2016 and 2025).

Preprocessing and balancing. We lower-case and validate domains and drop conflicts and duplicates, which leaves 999,664 legitimate and 3,170,189 DGA domains from 161 families of very different sizes (2 to 519,280 domains). Following Shahzad et al. [13], we strip the TLD and keep one domain per remaining name (e.g. `google.com` and `google.de` both give `google`), and drop the 15,086 names found in both classes. We then balance: we keep the families with at least 300 domains (64 smaller families are left out), sample at most 3,000 domains per family, and draw the same number of legitimate domains uniformly at random from the cleaned Tranco list. The remaining 10× family imbalance is handled with sample weights: legitimate and DGA weigh half each, and every DGA family weighs the same. Finally, we split the data 80/10/10 into training, validation and test sets, stratified by family; the test set is used only for the final results.

5 Proposed Models and Methods

Every model is trained in two setups on the same training set: *binary* (target DGA/legitimate) and *family-first* (target `legit` or one of the 97 families, collapsed with $P(\text{DGA}) = 1 - P(\texttt{legit})$). The neural models are also trained *multi-task* [2]: one shared encoder with a binary head and a family head whose losses are summed, so the family labels shape the latent space while the binary head makes the decision. Hyperparameters are chosen on the validation set.

Model 0: majority class. Always predicts the most frequent class. It is the floor every model must beat.

Model 1: logistic regression (scikit-learn [11]). *Why it suits the data:* it is fast and interpretable, and it is linear: a binary model draws a single boundary between legitimate names and all DGAs, although families sit in very different regions (random strings vs. dictionary words). *What we learn:* family-first gives it one boundary per family, so if family labels help anywhere, they should help most here.

Model 2: character-level CNN [6], in PyTorch [10]. It embeds each character and applies convolutions, learning local character patterns (hex runs, alternating vowels and consonants, word pieces) wherever they occur in the name. *Why it suits the data:* families differ mostly in such local patterns. *What we learn:* comparing it with Model 1 shows whether the gain from family labels shrinks as the model becomes more flexible, and the multi-task setup shows whether that gain can be kept without the collapse step.

Model 3: character-level Transformer encoder [15]. Self-attention relates characters anywhere in the name, not only neighbours. *Why it suits the data:* dictionary DGAs, the hardest case [14], are built from whole words, so their pattern spans the name. *What we learn:* whether a model that sees the whole name at once closes the gap on dictionary families, and how the three setups compare for it. Finally, we test whether the CNN or Transformer embeddings, given to the simple Model 1, are enough to match the full neural models.

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